Faculty of Science and Engineering

Department of Mathematics and Computer Science

ASSESSMENT BRIEF

Module Code	7CS033.
Module Leader	Amin Noroozi, Anirban Chakraborty
Semester	2
Year	2023-24.
Assessment	Portfolio.
% of module mark	100%.
Set Date	Teaching week 1.
Due Date	15 March 2024, before 2 pm
Hand-in – what?	The requirements for the portfolio are outlined below.
Hand-in- where?	Canvas
Pass mark	50% grade required to pass the module.
Method of retrieval	Please see the reassessment section for more information.
Feedback	Provisional grades and feedback will be available to students within 4 weeks of submission via Canvas.
Collection of marked work	Available within 4 weeks of the submission date via Canvas.
University Regulations	University's Academic Regulations: https://www.wlv.ac.uk/about-us/governance/legal-information/policies-and-regulations/academic-regulations/ Academic Integrity Policy: https://www.wlv.ac.uk/about-us/corporate-information/wlv-policies/academic-integrity-policy/ Performance descriptors are linked here: Level and Mark Descriptors University Performance Descriptor: https://smsweb.wlv.ac.uk/urd/sits.urd/stu_records/percentages.pdf
Support	Student Support and Wellbeing: https://www.wlv.ac.uk/current-students/student-support/student-support-and-wellbeing-ssw/ Study Guides: https://www.wlv.ac.uk/lib/skills-for-learning/study-guides/ Online Student Handbook: https://www.wlv.ac.uk/current-students/student-handbook/ Assessment information: https://www.wlv.ac.uk/current-students/student-handbook/your-course/assessments-and-grades/

OVERVIEW

Description	Each week, you will complete the tasks at the end of a Jupyter Notebook and then generate a PDF version of the same Jupyter Notebook. The tasks include coding and writing reports where requested.
Guidelines	Detailed guidelines for the assessment, including tasks, submission, and marking criteria, are listed below.
Self- assessment checklist Make sure that you…	• The required components are detailed below (Please see guidelines). Use them to structure your work. • Use the materials discussed in the lectures and tutorials to complete the tasks. • Your codes run correctly. • You upload the assignment on Canvas according to the formatting guidelines
Three key pieces of advice based on the feedback given to the previous cohort who completed this assignment	• Read the task description thoroughly and ask for clarification. • Read the formatting guidelines and stick to them. • Start working on your coursework early. Ensure that you understand the tasks and the submission process.
Submission Formatting Guidelines	• Your submission should be in the form of a single ZIP file named as follows: StudentName-StudentNumber-7CS033 • The ZIP file should contain 6 Jupyter Notebooks and the corresponding PDF files (One Jupyter Notebook and PDF file for each week). • You shouldn't submit the Jupyter Notebooks and PDF files weekly. You should wait until the last teaching week finishes and then submit all Jupyter Notebooks and their PDF versions in a ZIP file before the deadline (due date shown in the table above.
Referencing Style	Harvard referencing style (if needed)
Assessment Criteria and Marking Scheme	The assessment criteria and marking scheme are listed below.

LEARNING OUTCOMES

1. Understanding of the knowledge discovery process from data collection, processing to representation.
2. Understanding and usage of the concepts and main techniques in data mining.
3. Develop a data mining solution, using a state-of-the-art data mining tool.

RESOURCES

Lecture slides and workshop activities. However, when needed, you are expected to research and use different resources to complete the tasks.

ASSESSMENT DESCRIPTION

You are required to apply the data mining methods covered in the lectures to a number of tasks given to you in each workshop and prepare a portfolio with a 3000-word limit as follows:

Each week you will be given a Jupyter Notebook which has two sections. The first section includes some exercises and questions with solutions covering the materials taught in the lectures. You need to read through this section and complete the exercises. The exercises don't have any marks and are only for training and preparing you for the second section. The second section includes some marked tasks which comprise a part of the assessment and count toward your final mark in this module. You need to complete the tasks, generate a PDF version of the notebook, and submit them after week 6 according to the submission guidelines provided in the submission section.

For each task, you will see two empty cells with the titles "WRITE YOUR CODE IN THIS CELL" and "WRITE YOUR REPORT IN THIS CELL (IF APPLICABLE)", which will be referred to as "Code Cell" and "Report Cell", respectively, in this document. You should write your code in the code cell. If the task asks you to provide a report, or if you want to add some explanations, you need to write them in the report cell.

NOTE 1: If filling the report cell is required, it will be shown by the following statement at the end of the task description: "Completing the report cell is required". **In this case, you will lose marks if you don't complete the report cell**. You will not lose any marks if you leave the report cell empty for any task that doesn't include the statement mentioned above.

NOTE 2: You should write the number of the words used in the report cell in round brackets at the end of each report cell. **The total number of words used in all report cells of all workshops shouldn't exceed 3000 words**. The code cells do NOT count toward the 3000-word limit.

NOTE 3: You must run the code cell and print the output (where applicable) before submitting the notebook. **You will lose marks if you submit the notebook without running the code cell and printing its output**.

NOTE 4: You can create new cells if you need more than one cell for your code or report. If you need new cells, please add a comment to the top of the new cell and specify whether it is a code cell or a report cell. For example, add the following comment to the top of the cell:

This is a code cell for Task 1.1

OR

This is a report cell for Task 1.1

You can replace Task 1.1 with the task number for which you are writing your code.

NOTE 5: The tasks are based on the lecture materials and the exercise section of the notebook. To complete the tasks, you need to study the lecture files and complete the exercise section (the first section) of the notebook.

NOTE 6: The marks for the task are shown in the notebooks.

SUBMISSION

The final submission should include the Jupyter Notebooks and corresponding PDF files from all workshops compressed into a single ZIP file. This means the ZIP file should contain six Jupyter notebooks and six PDF files (one notebook and one PDF file for each week's workshop).

The ZIP file should be named as follows: StudentName-StudentNumber-7CS033.

DEADLINE

The deadline to submit the ZIP file is **15 March 2024, before 2 pm**.

NOTE 5: You shouldn't submit the notebooks and PDF files each week. You should put all files in a ZIP file as explained in the submission section and submit the ZIP file after week 6 and before the deadline mentioned above.

NOTE 6: Late submissions even one minute after the deadline will be treated according to the university's late submission and extension policy and procedure which can be found at the following link: Late submission and extension policy and procedure

REASSESSMENT (RESIT)

All criteria and specifications for the reassessment are the same as the assessment and students need to improve and complete their assessment submissions described in this document.

The reassessment time will be during the university semester 2 reassessment period.

The **deadline** to submit the reassessment portfolio will be **10 July 2024, before 2 pm**.

ASSESSMENT CRITERIA (MARKING SCHEME)

The following table shows the assessment criteria adapted from the University performance descriptor provided at the following link: University Performance Descriptor.

90-100% Outstanding	**Overview**: The marks of the correct solutions add up to 90-100%, however with a few minor errors, but without any major errors. **Explanation:** Exceptional work that demonstrates an excellent knowledge and understanding of complex issues and methodologies at the forefront of data mining and informatics. Contains exceptional evidence of original independent critical thinking that is based upon a sophisticated and rigorous argument. Is accurately supported by evidence derived from a wide range of source material including primary sources and current research.
80-89% Excellent	**Overview**: The marks of the correct solutions add up to 80-89%, however with a few minor errors, or, the marks of the correct solutions add up to 90-100%, but with some major errors **Explanation:** Outstanding work that demonstrates an excellent level of understanding of complex issues and methodologies at the forefront of data mining and informatics. Displays independent critical thought, is strong and sophisticated, with well organised argument. Is accurately supported by evidence derived from a wide range of source material including primary sources and current research.
70-79% Very good	**Overview**: The marks of the correct solutions add up to 70-79%, however with a few minor errors, or, the marks of the correct solutions add up to 80-89%, but with some major errors **Explanation:** Excellent work that demonstrates an excellent level of understanding of complex issues and methodologies at the forefront of data mining and informatics. Displays independent critical thought and a strong, organised argument. Is accurately supported by evidence derived from a wide range of source material including primary sources and current research.
60-69% Good	**Overview**: The marks of the correct solutions add up to 60-69%, however with a few minor errors, or, the marks of the correct solutions add up to 70-79%, but with some major errors **Explanation**: Very good work demonstrating good understanding of issues related to data mining and informatics, including some complex Issues. Displays a good and well organised argument and evaluation with the ability to critically evaluate competing arguments. Is accurately supported by evidence derived from a wide range of source material including primary sources and current research.
50-59% Pass	**Overview**: The marks of the correct solutions add up to 50-59%, however with many minor errors, or, the marks of the correct solutions add up to 60-69%, but with some major errors **Explanation**: Good work showing satisfactory grasp of main issues, sufficient awareness of data mining and informatics. Argument identified and some analysis of key issues, but with limited critical judgement. Demonstrates sufficient familiarity with a proportion of the basic reading but with minor errors and/or omissions of essential material.
40-49% Fail	**Overview**: The amount of work done is considerable, but the marks of the correct solutions don't add up to 50%, even by ignoring minor errors. The solutions are mostly wrong or show lack of awareness of the subject **Explanation**: Unsatisfactory work showing only limited grasp of some of the issues, poorly conceived and poorly directed to the question or task set, or with serious errors or omissions and limited awareness of the subject or practice. Shows some evidence of planning, although irrelevant/unrelated material or arguments are included. Familiarity with a proportion of the basic reading but with errors and/or omissions of essential material.
30-39% Fail	**Overview**: Some codes and cells run and some reports are presented, but the work quality is unsatisfactory. The marks of the correct solutions don't add up to 40%, even by ignoring minor errors. **Explanation:** Showing very limited grasp of some relevant issues and necessary material and/or skills, or with major errors, omissions or misconceptions, and with very limited awareness of data mining techniques.
0-29% Fail	**Overview**: No or a few cells run, the result are mostly incorrect, the reports are not presented or presented very poorly. The marks of the correct solutions don't add up to 30%, even by ignoring minor errors. **Explanation:** No or No work offered; or work that is totally irrelevant to the question or task set, fundamentally wrong or shows fragmentary evidence of familiarity with course material or awareness of data mining and informatics

GENERAL GUIDELIENS

Please consider the following points before completing the workshop exercises and tasks

1- BELIEVE IN YOURSELF! This module is designed to walk you through Data Mining and Informatics from zero to hero. Nothing is going to stop you from achieving this aim.

2- Google is the number one friend of any competent programmer. No one knows everything. If you face anything new, please first google it using suitable keywords. For example, if a task asks you to find the maximum value in a table, you can search Google using the following text 'How to find the maximum value in a table Python Pandas'. You will become a superhero in programming by knowing how to google the right keyword for different problems.

3- Use a trial-and-error approach when you want to analyse a code. For example, if you need to see what each part of a code is doing, remove them one by one, run the code and observe what has changed. You can improve your skills very fast and efficiently using this approach.

4- If you cannot complete a task but can show your knowledge of the subject by writing a report or some codes, you will still get some marks according to the marking scheme. **So, do NOT leave any task unchecked or empty**.

5- As many of you have no programming experience, it is normal if you find yourself struggling in the beginning. This problem will be resolved after a few hours of practice if you don't give up. Persistence combined with consistency and the ability to face ambiguity is the key to becoming a skilful data scientist.

GOOD LUCK 😊